

## LEUSHI, Russian Federation

WMO: 280640

Lat: 59.633N

Lon: 65.717E

Elev: 72

StdP: 100.46

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |       |     |       | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|-------|-----|-------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |       | 1%  |       |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB  | WS  | MCDB  | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)   | (l) | (m)   | (n)                   | (o)  | (p)   |
| 1             | -34.9      | -32.2 | -37.9                         | 0.1 | -34.9 | -35.1 | 0.1 | -32.2 | 6.9                   | -14.8 | 6.3 | -12.5 | 1.4                   | 320  | 0.606 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.0                    | 28.9            | 20.1 | 27.1 | 19.2 | 25.2 | 18.1 | 21.1                | 27.2 | 20.1 | 25.5 | 19.0 | 23.8 | 2.0                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.0                            | 13.9 | 24.2 | 18.0 | 13.0 | 23.1 | 16.9 | 12.2 | 21.9 | 61.6          | 27.3 | 58.0 | 25.5 | 54.3 | 23.9 | 26.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 8.1               | 6.5  | 5.9 | -37.6                      | 31.9 | 4.4                | 1.4 | -40.7                                              | 32.9 | -43.3      | 33.7 | -45.8      | 34.5 | -49.0      | 35.5 |    |    |
|                   |      |     | -37.6                      | 22.8 | 4.4                | 1.4 | -40.8                                              | 23.8 | -43.4      | 24.7 | -45.8      | 25.5 | -49.0      | 26.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                   |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|-------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                   |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 1.2    | -17.6 | -14.8 | -5.3  | 2.9   | 9.7   | 15.9  | 18.2  | 15.3  | 9.6   | 1.8   | -7.8  | -14.5 |
|                                                                   | DBStd       | 13.77  | 8.38  | 7.97  | 5.60  | 5.38  | 5.46  | 4.79  | 4.14  | 4.09  | 4.06  | 5.04  | 7.79  | 8.28  |
|                                                                   | HDD10.0     | 3934   | 854   | 693   | 474   | 218   | 75    | 8     | 1     | 3     | 56    | 259   | 533   | 759   |
|                                                                   | HDD18.3     | 6354   | 1113  | 926   | 733   | 464   | 271   | 102   | 55    | 113   | 264   | 513   | 783   | 1018  |
|                                                                   | CDD10.0     | 728    | 0     | 0     | 0     | 4     | 67    | 186   | 256   | 167   | 43    | 4     | 0     | 0     |
|                                                                   | CDD18.3     | 106    | 0     | 0     | 0     | 0     | 4     | 29    | 52    | 19    | 1     | 0     | 0     | 0     |
|                                                                   | CDH23.3     | 837    | 0     | 0     | 0     | 0     | 58    | 227   | 411   | 136   | 6     | 0     | 0     | 0     |
|                                                                   | CDH26.7     | 187    | 0     | 0     | 0     | 0     | 8     | 50    | 98    | 31    | 1     | 0     | 0     | 0     |
|                                                                   | WSAvg       | 2.3    | 1.9   | 1.9   | 2.5   | 2.6   | 2.8   | 2.5   | 2.3   | 2.2   | 2.2   | 2.4   | 2.2   | 1.9   |
| Precipitation                                                     | PrecAvg     | 488    | 23    | 17    | 18    | 26    | 38    | 61    | 76    | 68    | 62    | 39    | 33    | 24    |
|                                                                   | PrecMax     | 690    | 46    | 35    | 35    | 72    | 81    | 134   | 179   | 178   | 136   | 92    | 73    | 43    |
|                                                                   | PrecMin     | 316    | 7     | 3     | 2     | 6     | 8     | 5     | 10    | 6     | 12    | 16    | 4     | 2     |
|                                                                   | PrecStd     | 95     | 9     | 9     | 9     | 15    | 22    | 30    | 43    | 33    | 33    | 18    | 18    | 10    |
| Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 0.9   | 2.5   | 9.2   | 19.0  | 27.9  | 30.6  | 30.8  | 29.9  | 23.7  | 16.9  | 6.6   | 2.0   |
|                                                                   |             | MCWB   | 0.4   | 0.3   | 4.1   | 11.7  | 17.4  | 19.8  | 21.2  | 20.7  | 16.5  | 10.8  | 4.2   | 1.0   |
|                                                                   | 2%          | DB     | -1.5  | 0.6   | 5.7   | 15.1  | 24.8  | 28.1  | 29.2  | 26.8  | 20.4  | 13.4  | 4.8   | 0.2   |
|                                                                   |             | MCWB   | -2.2  | -0.7  | 1.6   | 9.4   | 15.3  | 19.4  | 20.7  | 19.5  | 14.9  | 9.0   | 3.2   | -0.7  |
|                                                                   | 5%          | DB     | -4.1  | -1.6  | 3.8   | 12.9  | 22.2  | 26.1  | 27.6  | 24.4  | 17.8  | 10.7  | 2.7   | -1.7  |
|                                                                   |             | MCWB   | -4.7  | -2.9  | 0.8   | 7.8   | 13.9  | 18.8  | 19.7  | 18.1  | 13.3  | 7.6   | 1.5   | -2.6  |
|                                                                   | 10%         | DB     | -6.0  | -3.9  | 2.1   | 10.6  | 19.1  | 23.9  | 25.7  | 22.2  | 15.5  | 8.6   | 0.9   | -4.0  |
|                                                                   |             | MCWB   | -6.7  | -5.3  | -0.3  | 6.1   | 12.2  | 17.3  | 18.7  | 17.1  | 12.1  | 6.5   | -0.1  | -4.8  |
| Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 0.5   | 0.7   | 4.4   | 12.7  | 17.8  | 21.9  | 22.6  | 21.8  | 18.5  | 11.3  | 4.9   | 1.1   |
|                                                                   |             | MCDB   | 0.9   | 2.0   | 8.5   | 17.3  | 25.5  | 27.4  | 29.3  | 28.2  | 22.3  | 15.1  | 6.2   | 1.9   |
|                                                                   | 2%          | WB     | -2.3  | -0.6  | 2.3   | 10.3  | 15.9  | 20.5  | 21.4  | 20.2  | 15.6  | 9.5   | 3.3   | -0.6  |
|                                                                   |             | MCDB   | -1.7  | 0.6   | 5.1   | 14.0  | 24.2  | 26.3  | 27.7  | 25.4  | 19.4  | 12.6  | 4.8   | 0.2   |
|                                                                   | 5%          | WB     | -4.7  | -2.7  | 1.0   | 8.2   | 14.4  | 19.2  | 20.5  | 18.9  | 13.8  | 7.9   | 1.6   | -2.4  |
|                                                                   |             | MCDB   | -4.2  | -1.6  | 3.2   | 12.2  | 20.9  | 24.6  | 26.1  | 23.2  | 16.8  | 10.5  | 2.6   | -1.6  |
|                                                                   | 10%         | WB     | -6.7  | -5.2  | 0.0   | 6.4   | 12.8  | 18.0  | 19.4  | 17.6  | 12.6  | 6.4   | 0.1   | -4.7  |
|                                                                   |             | MCDB   | -6.1  | -4.1  | 1.8   | 10.2  | 18.4  | 22.7  | 24.5  | 21.4  | 15.1  | 8.6   | 0.8   | -4.0  |
| Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 6.6   | 7.7   | 8.3   | 8.9   | 10.0  | 9.1   | 9.0   | 8.3   | 7.2   | 5.7   | 5.5   | 6.2   |
|                                                                   |             | MCDBR  | 6.8   | 7.0   | 9.0   | 11.9  | 14.2  | 12.2  | 11.9  | 11.5  | 10.6  | 8.9   | 5.5   | 5.7   |
|                                                                   |             | MCWBR  | 6.6   | 6.2   | 6.3   | 7.7   | 7.6   | 6.0   | 5.3   | 5.6   | 6.0   | 5.5   | 4.8   | 5.5   |
|                                                                   | 5% WB       | MCDBR  | 6.9   | 6.7   | 8.0   | 11.2  | 13.0  | 11.1  | 11.2  | 10.7  | 9.5   | 8.3   | 5.5   | 5.6   |
|                                                                   |             | MCWBR  | 6.7   | 6.2   | 6.0   | 7.7   | 7.3   | 6.0   | 5.4   | 5.6   | 6.1   | 5.6   | 5.0   | 5.4   |
| Clear-Sky Solar Irradiance                                        | taub        |        | 0.266 | 0.304 | 0.360 | 0.396 | 0.394 | 0.400 | 0.413 | 0.420 | 0.365 | 0.334 | 0.288 | 0.257 |
|                                                                   | taud        |        | 2.006 | 2.029 | 1.964 | 2.084 | 2.288 | 2.339 | 2.302 | 2.248 | 2.370 | 2.304 | 2.105 | 1.959 |
|                                                                   | Ebn at Noon |        | 483   | 668   | 738   | 793   | 832   | 833   | 809   | 767   | 755   | 650   | 458   | 337   |
|                                                                   | Edh at Noon |        | 50    | 85    | 125   | 133   | 117   | 114   | 116   | 113   | 85    | 66    | 45    | 35    |
| All-Sky Solar Radiation                                           | RadAvg      |        | 0.50  | 1.42  | 2.92  | 4.24  | 5.06  | 5.47  | 5.33  | 3.92  | 2.36  | 1.19  | 0.48  | 0.26  |
|                                                                   | RadStd      |        | 0.04  | 0.10  | 0.20  | 0.42  | 0.41  | 0.40  | 0.47  | 0.33  | 0.24  | 0.16  | 0.08  | 0.03  |

## Historical Trends

|                             | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page